Task 1: Biological Dataset Selection and Description

1. Dataset Title

Human Insulin (INS) mRNA Sequence

2. Source Database

NCBI Nucleotide Database (National Center for Biotechnology Information)

3. Dataset Details

- **Organism:** *Homo sapiens* (human)
- **Gene name:** INS (insulin)
- **Transcript variant:** 1
- **Molecule type:** mRNA
- **Accession number:** NM_000207.3
- **Sequence length:** 465 nucleotides
- **Sequence format:** FASTA

4. Dataset Description

The selected dataset contains the human insulin (INS) messenger RNA sequence. The INS gene encodes insulin, a peptide hormone involved in regulating glucose and other aspects of metabolism. The selected transcript is publicly available in the NCBI Nucleotide database and can be identified using its accession number, NM_000207.3.

5. Reason for Dataset Selection

This dataset was selected because it is publicly accessible, has a clearly defined accession number, and represents a well-characterized human gene. It is suitable for learning basic bioinformatics tasks such as nucleotide sequence retrieval, sequence organization, and introductory sequence analysis.

6. Data Retrieval Method

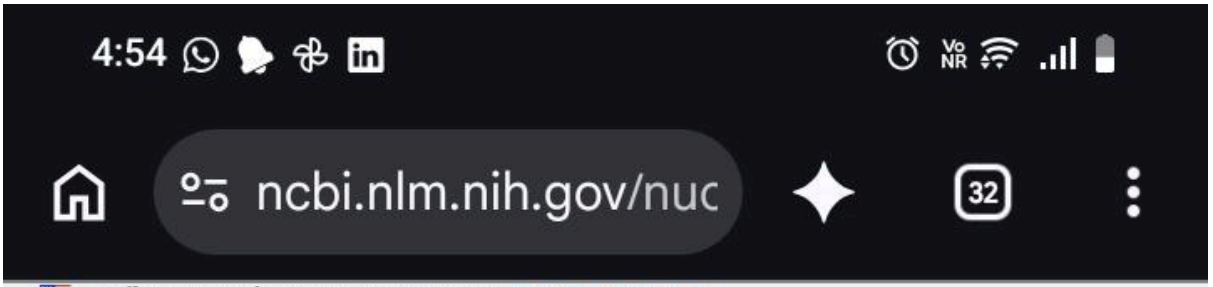
The dataset was retrieved from the NCBI Nucleotide database. The transcript record was opened, the FASTA format was selected, and the nucleotide sequence was copied. A screenshot was taken as evidence of the retrieval process.

7. Conclusion

The human INS mRNA sequence was successfully identified and retrieved from NCBI. Its accession number, organism, transcript type, and sequence length were recorded for the biological dataset selection task.

Reference

NCBI Nucleotide: https://www.ncbi.nlm.nih.gov/nuccore/NM_000207.3



An official website of the United States government Here's how you know

National Library of Medicine
National Center for Biotechnology Information

NIH Websites Are Changing

- Get the Latest Updates and FAQs on the transition
- Per Executive Order 14434, federal agencies are now using "Super Intelligence" (SI) in place of "Artificial Intelligence" (A terms across NIH content during this transition.

Nucleotide | Nucleotide | Advanced

FASTA | Send to: | Change | Custom

Homo sapiens insulin (INS), transcript variant 1, mRNA

NCBI Reference Sequence: NM_000207.3

GenBank Graphics

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>NM_000207.3 Homo sapiens insulin (INS), transcript variant 1, mRNA
AGCCCTCCAGGACAGGCTGCATCAGAAGAGGCCATCAAGCAGATCACTGTCCTTCTGCCATGGCCCTGTG
GATGCGCCTCCTGCCCCTGCTGGCGCTGCTGGCCCTCTGGGGACCTGACCCAGCCGCAGCCTTTGTGAAC
CAACACCTGTGCGGCTCACACCTGGTGGAAGCTCTCTACCTAGTGTGCGGGGAACGAGGCTTCTTCTACA
CACCCAAGACCCGCCGGGAGGCAGAGGACCTGCAGGTGGGGCAGGTGGAGCTGGGCGGGGGCCCTGGTGC
AGGCAGCCTGCAGCCCTTGGCCCTGGAGGGGTCCCTGCAGAAGCGTGGCATTGTGGAACAATGCTGTACC
AGCATCTGCTCCCTCTACCAGCTGGAGAACTACTGCAACTAGACGCAGCCCGCAGGCAGCCCCACACCCG
CCGCCTCCTGCACCGAGAGAGATGGAATAAAGCCCTTGAACCAGC
```

Analyze
Run BLA
Pick Prim
Show in

Referenc
RefSeq a
See 4 ref
variants

More ab
This gen
hormone
regulatio
metaboli
Also Kno
ILP...

Related
Protein
PubMed
Taxonom
Annotate
CCDS
Gene
OMIM
PubMed